Lesson 9.5 - Finding a Confidence Interval of a Population Proportion

> **Video:** Finding The Confidence Interval of a Population Proportion Using The Normal Distribution by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

Formulas

Quantity	Formula
Sample proportion ("p-hat" or "p-prime")	$\hat{p} = \frac{x}{n}$, where x = number of "yes" (successes)
Proportion of failures	$\hat{q} = 1 - \hat{p}$
Normal approximation is OK if	$n\hat{p}\hat{q} \geq 10$ (many texts check $n\hat{p} \geq 10$ and $n\hat{q} \geq 10$)
Error bound for a proportion (EBP)	$E = z\sqrt{\frac{\hat{p}\hat{q}}{n}}$
Confidence interval	$\hat{p} - E < p < \hat{p} + E$
Sample size needed	$n = \frac{z^2\,\hat{p}\,\hat{q}}{E^2}$, rounded **up**
Critical z	look up the left area $\frac{1 + \mathrm{CL}}{2}$

Common z values: 90% → 1.645, 95% → 1.96, 98% → 2.33, 99% → 2.575.

Example 1 - laptop ownership

800 students were surveyed and 584 own a laptop.

(a) Can the normal distribution be used? This is a binomial situation (yes/no answers).

$$\hat{p} = \frac{584}{800} = 0.73, \quad \hat{q} = 0.27, \quad n\hat{p}\hat{q} = 800(0.73)(0.27) = 157.68 \geq 10 \checkmark$$

(b) 95% confidence interval and margin of error ($z = 1.96$, since the left area is 0.975)

$$E = 1.96\sqrt{\frac{0.73 \times 0.27}{800}} = 1.96\sqrt{2.46375 \times 10^{-4}} \approx 0.0308$$

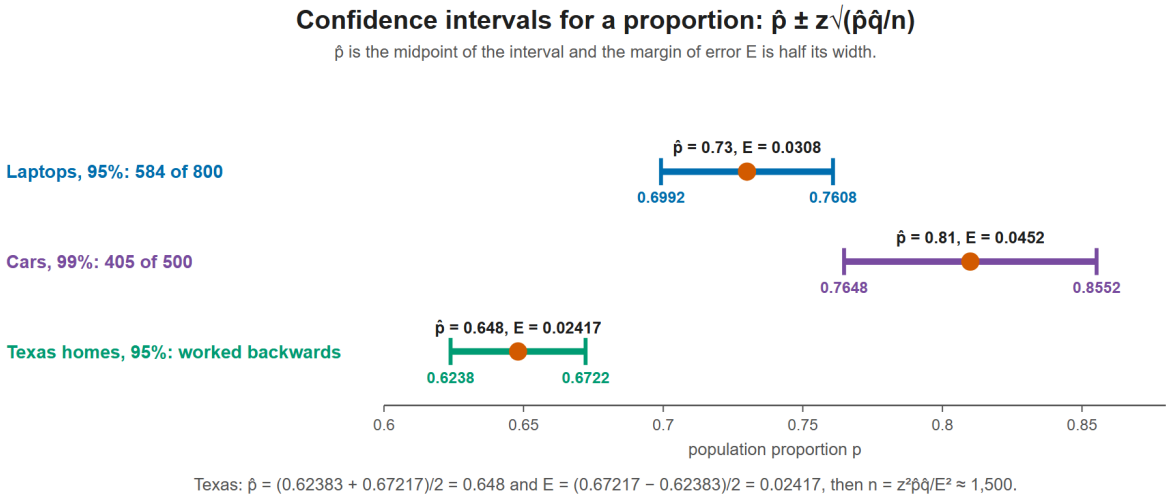
$$0.73 - 0.0308 < p < 0.73 + 0.0308 \;\Rightarrow\; 0.6992 < p < 0.7608$$

We are 95% confident that between about **69.9% and 76.1%** of all students own a laptop.

(c) Sample size for a 2% margin of error at 90% confidence ($z = 1.645$)

$$n = \frac{(1.645)^2(0.73)(0.27)}{(0.02)^2} \approx 1333.39 \;\Rightarrow\; n = 1334$$

Derivation: square $E = z\sqrt{\hat{p}\hat{q}/n}$ to get $E^2 = z^2\hat{p}\hat{q}/n$, then solve for n.



Texas: p̂ = (0.62383 + 0.67217)/2 = 0.648 and E = (0.67217 − 0.62383)/2 = 0.02417, then n = z²p̂q̂/E² ≈ 1,500.

The three intervals from the lesson. The dot is pÌ□, the midpoint, and each arm is the margin of error E. The 99% car interval is the widest, because higher confidence uses a larger z.

Example 2 - car ownership

500 residents were surveyed and 405 own a car.

$$\hat{p} = \frac{405}{500} = 0.81, \qquad \hat{q} = 0.19$$

(a) 99% confidence interval ($z = 2.575$, since the left area 0.995 lies between $z = 2.57$ and 2.58)

$$E = 2.575\sqrt{\frac{0.81 \times 0.19}{500}} = 2.575(0.01754) \approx 0.0452$$

$$0.7648 < p < 0.8552$$

(b) Sample size for a 3% margin of error at 98% confidence ($z = 2.33$, left area 0.99)

$$n = \frac{(2.33)^2(0.81)(0.19)}{(0.03)^2} \approx 928.34 \;\Rightarrow\; n = 929$$

Example 3 - working backwards from an interval

A 95% confidence interval for the proportion of homeowners in Texas is $(0.62383,\ 0.67217)$.

(a) Sample proportion. It is the **midpoint** of the interval:

$$\hat{p} = \frac{0.62383 + 0.67217}{2} = 0.648$$

(b) Error bound. It is **half the width** of the interval:

$$E = \frac{0.67217 - 0.62383}{2} = 0.02417 \quad (= 0.67217 - 0.648)$$

(c) How many people were surveyed?

$$n = \frac{(1.96)^2(0.648)(0.352)}{(0.02417)^2} \approx 1499.95 \;\Rightarrow\; n \approx 1500$$

(d) How many said yes?

$$x = n\hat{p} = 1500(0.648) = 972$$

Key takeaways

- $\hat{p} = x/n$, $\hat{q} = 1 - \hat{p}$. Check $n\hat{p}\hat{q} \geq 10$ before using the normal approximation.
- CI = $\hat{p} \pm z\sqrt{\hat{p}\hat{q}/n}$.
- Midpoint of a CI = $\hat{p}$. Half-width = margin of error.
- Sample size: $n = z^2\hat{p}\hat{q}/E^2$, always rounded **up**.